# Unit 1

## Short Answers

1. **What do you mean by Data Cleaning?**

Data cleaning, also known as data cleansing, is the process of identifying and correcting errors, inconsistencies, and inaccuracies in a dataset to improve its quality and reliability for analysis. It involves tasks such as removing duplicate records, handling missing values, correcting typos, and standardizing data formats.

2. **What are the limitations of Data mining?**

**Limitations of Data Mining:**

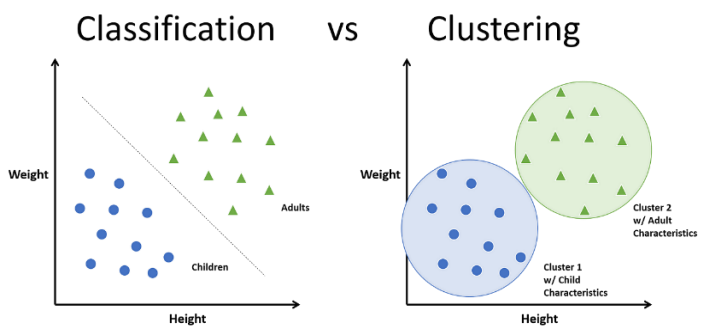
1. **Data Quality:** Data mining outcomes heavily rely on the quality of input data. Poor-quality or noisy data can lead to inaccurate results.
2. **Overfitting:** Complex models may capture noise or irrelevant patterns in the data, leading to overfitting and poor generalization to unseen data.
3. **Interpretability:** Some data mining models, especially black-box algorithms like neural networks, lack interpretability, making it challenging to understand and trust their predictions.
4. **Computational Complexity:** Analyzing large datasets using sophisticated data mining techniques can be computationally intensive and time-consuming.

3. **What is classification?**

Classification is a supervised learning technique in machine learning where the goal is to categorize input data into predefined classes or categories based on their features. It involves training a model on labeled data to make predictions about the class labels of new, unseen instances.

4. **Differentiate between classification and clustering:**

| Classification | Clustering |
|---|---|
| Model predicts a probable classification for a given input | Model maps a given input into one of the data clusters |
| Uses supervised learning | Uses unsupervised learning |
| Labeled data required for training | Does not require labeled data |
| Number of categories is known | Number of groups is unknown |
| Two-step process to train and predict | Single-step process |




5. **Can you clarify the difference between Structured and Unstructured data? How are they relevant to Big Data?**

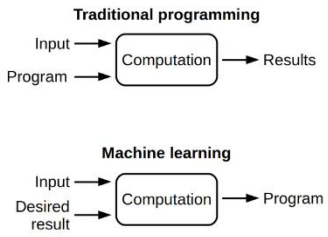
- **Structured Data:** Structured data refers to data that is organized and formatted in a specific way, typically with a clear schema. It is easily searchable, queryable, and analyzable using traditional database systems. Examples include relational databases, spreadsheets, and CSV files.

| Name | Class | Section | Roll No | Grade |
|---|---|---|---|---|
| Geek1 | 11 | A | 1 | A |
| Geek2 | 11 | A | 2 | B |
| Geek3 | 11 | A | 3 | A |

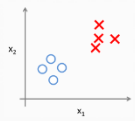
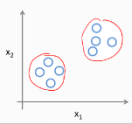
- **Unstructured Data:** Unstructured data lacks a predefined data model or structure. It includes text documents, images, audio files, videos, social media posts, and sensor data. Analyzing unstructured data requires advanced techniques such as natural language processing, image recognition, and sentiment analysis.
- **Relevance to Big Data:** Both structured and unstructured data are relevant to Big Data. Big Data refers to datasets that are too large, diverse, or complex to be processed using traditional data processing applications. Big Data technologies, such as Hadoop and Spark, are designed to handle massive volumes of both structured and unstructured data, enabling organizations to extract valuable insights and make data-driven decisions at scale.

---

1. **What is Data Mining and list few techniques of Data Mining:**
    - **Data Mining:** Data mining is the process of discovering patterns, correlations, or relationships within large datasets to extract useful information and make predictions. It involves various techniques to analyze and interpret data.
    - **Techniques of Data Mining:**
        1. **Classification:** It categorizes data into predefined classes or labels based on input features. Example: Decision Trees, Naive Bayes Classifier.
        2. **Clustering:** It groups similar data points together based on their characteristics or attributes. Example: K-means Clustering, Hierarchical Clustering.
        3. **Association Rule Mining:** It identifies relationships or associations among a set of items in a dataset. Example: Apriori Algorithm, FP-Growth Algorithm.
        4. **Regression Analysis:** It predicts the value of a dependent variable based on one or more independent variables. Example: Linear Regression, Polynomial Regression.
2. **Define Hash Functions and Natural Logarithms:**
    - **Hash Functions:** Hash functions are algorithms that map data of arbitrary size to fixed-size values. These functions are commonly used in data structures like hash tables to quickly locate a data record given its search key.
    - **Natural Logarithms:** Natural logarithms, denoted by "ln," are logarithms with base e, where e is an irrational constant approximately equal to 2.71828. They represent the inverse operation of exponentiation. Natural logarithms find applications in various mathematical and scientific contexts.
3. **What is machine learning:**
    - **Machine Learning:** Machine learning is a subset of artificial intelligence that enables systems to learn and improve from experience without being explicitly programmed. It involves developing algorithms and models that learn patterns and make predictions or decisions based on input data.




4. **Differentiate Supervised and Unsupervised learning:**

| Supervised | Unsupervised |
|---|---|
| Input Data is labelled | Input Data is Unlabelled |
| Uses training Dataset | Uses just input dataset |
| Data is classified based on training dataset | Uses properties of given data to classify it. |
| Used for prediction | Used for Analysis |
| Divided into two types Regression & Classification | Divided into two types Clustering & Association |
| Known number of classes | Unknown number of classes |
|  |  |
| Use off-line analysis of data | Use Real-Time analysis of data |

**5. Define Feature Extraction and list any 2 feature extraction techniques:**

- **Feature Extraction:** Feature extraction is the process of transforming raw data into a reduced set of relevant features that facilitate efficient analysis and modeling while retaining essential information.
- **Feature Extraction Techniques:**
    1. **Principal Component Analysis (PCA):** PCA identifies the orthogonal axes that capture the maximum variance in the data and projects the original data onto these axes.
    2. **Linear Discriminant Analysis (LDA):** LDA finds the linear combinations of features that best separate classes in labeled data, maximizing the between-class scatter and minimizing the within-class scatter.

---

**Unit 2**

Short Questions:

1. **Define Confidence and Support:**
    - Confidence measures the likelihood that a rule is true given that another rule is true. Support quantifies the frequency of occurrence of items in a dataset.
2. **Define Frequent Itemset, Maximal Frequent Itemset, and Closed Frequent Itemset:**
    - Frequent Itemset: A set of items that appear together frequently in a dataset.
    - Maximal Frequent Itemset: A frequent itemset that is not a subset of any other frequent itemset.

- Closed Frequent Itemset: A frequent itemset for which no proper super-set has the same support.

3. **What are the different ways of improving the efficiency of the Apriori algorithm?**
    - Pruning strategies, such as the use of hash-based techniques or reducing the size of candidate itemsets.
    - Parallelizing computation across multiple processing units or machines.
    - Utilizing specialized data structures like FP-trees to avoid candidate generation.
4. **What are the three major components of the Apriori algorithm in data mining?**
    - Candidate Generation: Creating potential itemsets based on the frequent itemsets found in the previous iteration.
    - Support Counting: Counting the occurrences of candidate itemsets in the dataset to determine their support.
    - Pruning: Removing candidate itemsets with support below the minimum threshold.
5. **What is FP-tree and FP Growth algorithm?**
    - FP-tree: A data structure used to efficiently store and process frequent itemsets in the FP Growth algorithm.
    - FP Growth algorithm: A method for mining frequent itemsets by constructing an FP-tree and recursively mining frequent itemsets from it.
6. **Define Association Rules for data mining.**
    - Association Rules describe relationships between items in a dataset, often in the form of "if-then" statements, indicating that the occurrence of one item implies the occurrence of another.

**1. Mention the importance of Association Rule Mining?**

- Association Rule Mining is crucial for discovering relationships and patterns in large datasets. It helps in understanding the co-occurrence of items or events, aiding in decision-making, product recommendation, and market basket analysis.

**2. List out the applications of Association rules**

- Market basket analysis
- Recommender systems
- Cross-selling and upselling strategies
- Web usage mining
- Healthcare data analysis
- Fraud detection
- DNA sequence analysis

**3. Define frequent sets, confidence and support**

- Frequent sets: Sets of items or events that occur together frequently in a dataset.
- Support: The frequency of occurrence of a set of items in the dataset, measured as the proportion of transactions containing that set.
- Confidence: The conditional probability that a certain set of items occur together given the occurrence of another set of items in the same transaction.

**4. Explain market basket analysis with an example**

- Market basket analysis examines the purchase behavior of customers to identify associations between products. For example, if customers who buy bread often also purchase butter, a market basket analysis can reveal this association. Retailers can then use this insight to optimize product placement or create targeted promotions.

**5. Define quantitative association rule**

- A quantitative association rule extends traditional association rules by incorporating quantitative measures such as numeric values or ranges. For example, "If the temperature exceeds 30°C, ice cream sales increase by 20%."

**6. Explain join step and prune step in Apriori algorithm**

- Join step: In the Apriori algorithm, the join step involves creating candidate itemsets of higher lengths by joining frequent itemsets of lower lengths. For example, if {A} and {B} are frequent itemsets, the join step combines them to form {A, B}.

- Prune step: In the Apriori algorithm, the prune step eliminates candidate itemsets that are infrequent by applying the Apriori property. This property states that if any subset of a candidate itemset is infrequent, the candidate itemset itself is also infrequent. This helps reduce the number of candidate itemsets to be considered in subsequent iterations, improving efficiency.

**7. What is the difference between multistage and multihash algorithm?**

- Multistage algorithm: Involves breaking down a complex task into multiple stages or phases, with each stage performing a specific function. This approach allows for better management of resources and optimization of performance.

- Multihash algorithm: Utilizes multiple hash functions to map data elements to hash values, improving the distribution of data across hash buckets and reducing collisions. This enhances the efficiency of data retrieval and manipulation operations.

---

## Unit 3

### Short Answers

1. **Strengths of Hierarchical Clustering:**

- Captures hierarchical structure: Hierarchical clustering reveals the hierarchical relationships between data points, allowing for a deeper understanding of the data's organization.
- No need to specify the number of clusters: Unlike partitioning methods, hierarchical clustering does not require the number of clusters to be predefined, making it more flexible.
- Interpretability: The dendrogram generated in hierarchical clustering provides a visual representation of the clustering process, making it easier to interpret and analyze.
- Handles various cluster shapes: Hierarchical clustering can accommodate clusters of different shapes and sizes, making it suitable for datasets with complex structures.
- Can be used for data exploration: Hierarchical clustering can be used as an exploratory tool to uncover patterns and relationships within the data, aiding in hypothesis generation and feature selection.

2. **Comparison between Agglomerative and Divisive Methods:**

| Aspect | Agglomerative Method | Divisive Method |
|---|---|---|
| Approach | Bottom-up approach, starts with individual data points as clusters and merges them iteratively | Top-down approach, starts with all data points in one cluster and splits them iteratively |

| Aspect | Agglomerative Method | Divisive Method |
|---|---|---|
| Complexity | Generally slower and computationally intensive due to merging step | Can also be computationally intensive, especially with large datasets |
| Interpretability | Provides a hierarchical structure (dendrogram) for visualization | Results in a flat clustering structure without hierarchy |
| Initialization | Typically begins with each data point as a singleton cluster | Begins with all data points grouped into a single cluster |
| Cluster Shape Handling | May struggle with non-globular or irregularly shaped clusters | May effectively handle clusters of various shapes and sizes |

3. **Outlier Detection using Data Mining:**

- Distance-based methods: Outliers are detected based on their distance from other data points. Points that fall significantly outside the range of neighboring points are considered outliers.
- Density-based methods: Outliers are identified by their low density in high-density regions or their isolation from dense regions. DBSCAN is a popular density-based method for outlier detection.
- Clustering-based methods: Outliers are points that do not belong to any cluster or belong to small, sparse clusters. They are detected during the clustering process as points with insufficient nearby neighbors.
- Statistical methods: Outliers are identified based on statistical properties such as mean, median, and standard deviation. Points that deviate significantly from the statistical norms are considered outliers.

4. **Strengths of Hierarchical Clustering:**

- Captures hierarchical structure
- No need to specify the number of clusters
- Interpretability
- Handles various cluster shapes
- Can be used for data exploration

**5. Consider a cluster C1={9,12,10,8,11} then calculate Clustering Feature(CF)?**

In the BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) algorithm, the Clustering Feature (CF) for a cluster is typically represented as a tuple containing three values:

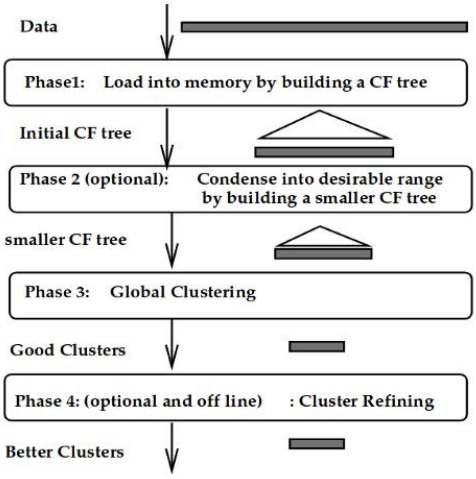
1. **Count (N):** The number of data points in the cluster.
2. **Linear Sum (LS):** The sum of all data points in each dimension.
3. **Squared Sum (SS):** The sum of squares of all data points in each dimension.

To calculate the CF for the given cluster $C1 = \{9, 12, 10, 8, 11\}$:

1. **Count (N):** Since there are 5 data points in the cluster, $N = 5$.
2. **Linear Sum (LS):** Calculate the sum of each dimension separately:
   - For dimension 1: $LS_{d1} = 9 + 12 + 10 + 8 + 11 = 50$
3. **Squared Sum (SS):** Calculate the sum of squares for each dimension separately:
   - For dimension 1: $SS_{d1} = 9^2 + 12^2 + 10^2 + 8^2 + 11^2 = 485$

So, the Clustering Feature (CF) for the cluster $C1$ is $(N = 5, LS_{d1} = 50, SS_{d1} = 485)$.

**7. What are the different phases of BIRCH how they are important in clustering?**




These phases are essential in the BIRCH algorithm for several reasons:

- **Efficiency**: By initializing with a small sample of the data, the algorithm can efficiently process large datasets without having to hold the entire dataset in memory at once.
- **Robustness**: Moving representative points helps adjust the clustering to account for variations in cluster shapes and sizes, making the algorithm more robust.
- **Scalability**: The merging phase allows the algorithm to scale to large datasets by gradually reducing the number of clusters.
- **Accuracy**: By iteratively refining the clustering through merging and point assignment, the algorithm aims to produce accurate clustering results even in the presence of noise or outliers.

**9. Concept of centroid linking in hierarchical clustering.**

Centroid linkage is a method used in hierarchical clustering to measure the distance between two clusters. In centroid linkage, the distance between two clusters is calculated as the Euclidean distance between their centroids, which are the mean vectors of the data points in each cluster.

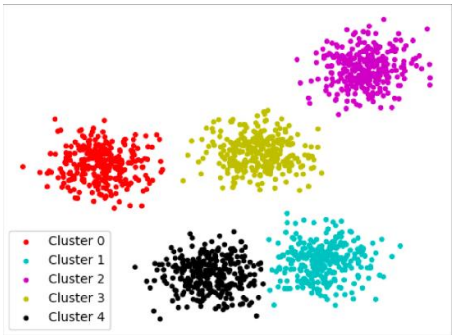
Here's how centroid linkage works:

1. **Calculate Centroids**: First, for each cluster, compute the centroid by finding the mean vector of all the data points in that cluster. The centroid represents the "center" of the cluster in the feature space.
2. **Compute Distance**: Once centroids are computed for each cluster, the distance between two clusters is calculated as the Euclidean distance between their centroids. This distance represents how far apart the centers of the two clusters are in the feature space.
3. **Merge Closest Clusters**: In hierarchical clustering, the two closest clusters are merged at each step. Centroid linkage determines the distance between clusters and identifies the pair with the smallest distance, indicating that they are the most similar and should be merged.
4. **Repeat**: The merging process continues iteratively until all data points are in a single cluster or until a stopping criterion is met.

---

1. **What is clustering? Why do businesses need to do clustering?**

- Clustering in data mining groups similar data points together based on their features and characteristics, forming clusters of similar objects.
- It helps identify patterns and similarities within datasets, aiding in decision-making processes.

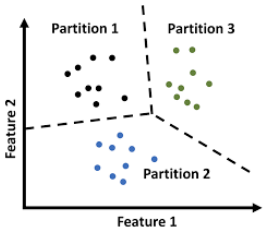



**Clustering is crucial for businesses because it enables:**

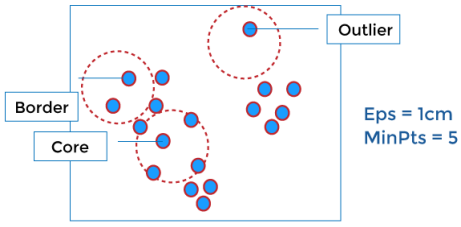
- Market Segmentation: Tailoring marketing strategies to specific customer segments based on their preferences and behaviors.
- Customer Segmentation and Targeting: Identifying high-value customer segments for targeted marketing campaigns.
- Product Segmentation and Customization: Categorizing products based on features and feedback to offer personalized product recommendations.
- Anomaly Detection: Identifying outliers or unusual patterns in data for fraud detection or risk management.
- Resource Allocation and Optimization: Optimizing resource allocation and operational efficiency by clustering similar resources.
- Pattern Recognition and Trend Analysis: Uncovering hidden insights and relationships within datasets to inform strategic decision-making.

2. **What are the major clustering methods?**

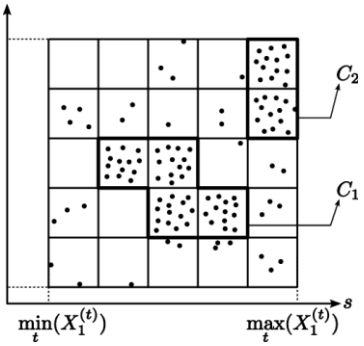
- **Partitioning Method:** Divides data into partitions forming clusters, using techniques like K-Means or K-Medoids.



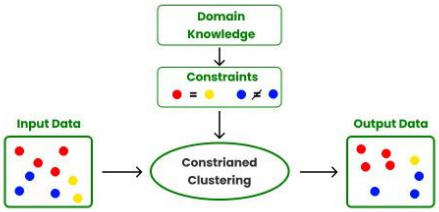

- **Hierarchical Method:** Creates a hierarchical decomposition of data, either agglomerative (bottom-up) or divisive (top-down).
- **Density-based Method:** Focuses on density to form clusters, such as DBSCAN, suitable for datasets with varying densities.




- **Grid-Based Method:** Quantizes data space into a grid structure and forms clusters based on grid cells' contents.




- **Model-Based Method:** Hypothesizes clusters based on a statistical model, like Gaussian Mixture Models (GMM).
- **Constraint-Based Method:** Incorporates user-defined constraints or application requirements into the clustering process to guide cluster formation.

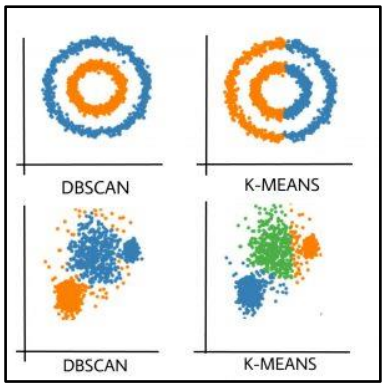



3. **When Should We Use DBSCAN Over K-Means In Clustering Analysis?**

*DBSCAN(Density-Based Spatial Clustering of Applications with Noise)* and K-Means are both clustering algorithms that group together data that have the same characteristic. However, They work on different principles and are suitable for different types of data. We prefer to use DBSCAN when the data is not spherical in shape or the number of classes is not known beforehand.

**Difference Between DBSCAN and K-Means.**

| DBSCAN | K-Means |
|---|---|
| In DBSCAN we need not specify the number of clusters. | K-Means is very sensitive to the number of clusters so it need to specified |
| Clusters formed in DBSCAN can be of any arbitrary shape. | Clusters formed in K-Means are spherical or convex in shape |
| DBSCAN can work well with datasets having noise and outliers | K-Means does not work well with outliers data. Outliers can skew the clusters in K-Means to a very large extent. |
| In DBSCAN two parameters are required for training the Model | In K-Means only one parameter is required is for training the model |




4. **Define Clustering Features in BIRCH Algorithm:**

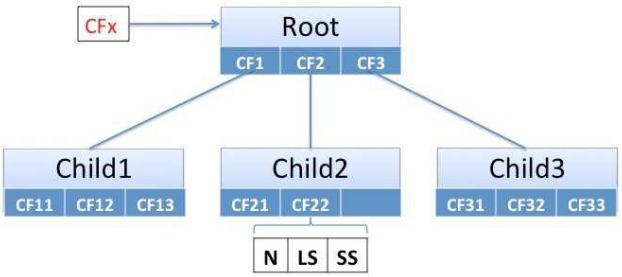
In the BIRCH (Balanced Iterative Reducing and Clustering using Hierarchies) algorithm, clustering features are defined based on two main components: CF (Clustering Feature) entries and CF Tree.

- **Clustering Feature (CF) Entry:** Compact representations of clusters within the dataset, storing aggregated information like cluster center, radius, number of points, and sum of squared deviations.
    1. **Cluster Center:** The centroid or mean of the data points belonging to the cluster.
    2. **Cluster Radius:** The radius or diameter of the cluster, representing the spread or compactness of the cluster.
    3. **Number of Points:** The count of data points belonging to the cluster.
    4. **Squared Sum of Deviations:** The sum of squared distances of data points from the cluster center, used for calculating the cluster radius.
- **CF Tree:** A hierarchical data structure organizing CF entries to facilitate efficient clustering, with nodes representing CF entries and branches representing relationships between clusters.

Clustering features enable BIRCH to dynamically and incrementally cluster incoming data points while maintaining the clustering quality within specified resource constraints.

**5. What is CF tree in BIRCH algorithm**

- The CF Tree is a hierarchical data structure used to organize CF entries and facilitate efficient clustering. It consists of nodes representing CF entries and branches representing the relationships between clusters at different levels of granularity.
- At each level of the CF Tree, CF entries are organized into clusters based on their proximity and similarity. Clusters are recursively merged or split based on predefined criteria to ensure the balance and efficiency of the clustering process.
- The CF Tree allows for fast and scalable clustering by efficiently managing and traversing the hierarchical clustering structure.




---

**Unit 4**

**Short answers**

**1. Define Mining data streams? Give an example of it.**

- Mining data streams involves extracting knowledge in real-time from continuous and infinite streams of volatile data.
- Unlike traditional databases, where data is stored for repeated access, data streams do not have a defined endpoint and continuously evolve over time.
- Techniques such as approximation and compression are used to handle the high volume and velocity of data streams efficiently.
- For example, monitoring social media feeds in real-time to detect emerging trends or analyzing sensor data from IoT devices to predict equipment failures are instances of mining data streams.

**2. What are the applications of stream data model?**

Stream data models find diverse applications across various domains due to their ability to process and analyze real-time data streams efficiently. Some common applications include:

- **Real-time Analytics**: Analyzing data streams for immediate insights, such as monitoring website traffic, social media sentiment analysis, or stock market analysis.
- **Network Traffic Monitoring**: Detecting anomalies and security threats in network traffic to prevent cyber attacks or network failures.
- **IoT Data Processing**: Processing sensor data from IoT devices to monitor and control physical systems, such as smart cities, industrial machinery, or environmental monitoring.
- **Fraud Detection**: Identifying fraudulent activities in financial transactions or online transactions in real-time to prevent financial losses.
- **Healthcare Monitoring**: Monitoring patient vital signs or medical device data streams for early detection of health issues or anomalies.
- **Supply Chain Management**: Tracking and analyzing data streams related to inventory, logistics, and production processes to optimize supply chain operations and improve efficiency.

**5. What is stream computing?**

Stream computing refers to the processing and analysis of continuous data streams in real-time or near-real-time. Unlike traditional batch processing, where data is collected and processed in batches, stream computing handles data as it is generated, allowing for immediate analysis and response. Stream computing systems are designed to handle high-volume, high-velocity data streams efficiently, often using parallel processing and distributed computing architectures. These systems are commonly used in applications such as real-time analytics, monitoring, and decision-making in various domains including finance, telecommunications, healthcare, and Internet of Things (IoT) systems.

**6. Write a note on page rank.**

- PageRank is an algorithm that was designed for the initial implementation of the Google search engine.
- The PageRank is defined as an approach for computing the importance of the web pages in a big web graph.
- PageRank – Basic Concept: A page or a node in a graph is as important as the number of links it has. Not all in-links are equal kind of links. The links coming from important pages are worth more. The importance of a page depends on the importance of other pages that points to it.
- The algorithm considers both the quantity and quality of links, with links from highly-ranked pages carrying more weight. PageRank is one of the key factors used by search engines to determine the relevance and importance of web pages.

**7. What is DGIM method?**

- DGIM algorithm (*Datar-Gionis-Indyk-Motwani Algorithm)*

- Designed to find the number 1's in a data set. This algorithm uses $O(\log^2 N)$ bits to represent a window of N bit, allows to estimate the number of 1's in the window with and error of no more than 50%.
- So this algorithm gives a 50% precise answer.
- In DGIM algorithm, each bit that arrives has a timestamp, for the position at which it arrives. if the first bit has a timestamp 1, the second bit has a timestamp 2 and so on.. the positions are recognized with the window size N (the window sizes are usually taken as a multiple of 2).The windows are divided into buckets consisting of 1's and 0's.

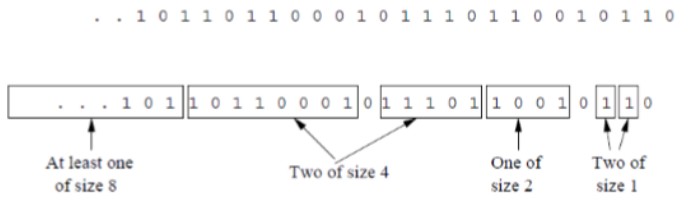



Figure: A bit-stream divided into buckets following the DGIM rules

**8. What is link analysis?**

- Link analysis is a data analysis technique used in network theory to analyze the links of the web graph.
- Graphs consists of a set of nodes and a set of edges or connections or intersections between nodes.
- The graphs are used everywhere, for example, in social media networks such as Facebook, Twitter, etc.

**9. What is the term spam?**

Term spam refers to the practice of artificially inflating the relevance of a web page or document to specific search terms or keywords in order to manipulate search engine rankings and attract more traffic to the page.

Term spam is a technique where somebody will run a query for the word e.g., theatre, but some other web page (e.g., T-shirt webpage) will appear with high rank.

---

**1. Define the characteristics of data streams**

Characteristics of data streams:

1. **Continuous and Infinite**: Data streams flow continuously in real-time, without a defined endpoint.
2. **Volatile and Evolving**: Data within streams is volatile, constantly changing and updating over time.
3. **Limited Resources**: Due to high volume, resources for storing and processing data streams are constrained.
4. **Real-Time Processing**: Data stream analysis occurs in real-time or near-real-time, requiring efficient algorithms.
5. **Approximation and Compression**: Techniques are used to compress and approximate data for efficient querying.
6. **Dynamic Querying**: Queries must adapt dynamically to handle changing data streams.
7. **High Velocity**: Data arrives rapidly in data streams, requiring fast processing.
8. **Variety of Data Types**: Data streams contain diverse data types, necessitating flexible processing techniques.

---

**Unit 5**

**Short Answers**

1. **Why is mining a social network graph important?**
    - Understanding social network structures helps identify influential users, detect communities, and analyze information diffusion, crucial for marketing, research, and social sciences.
2. **What is a social graph?**

- A social graph is a representation of social relationships among individuals or entities in a network, where nodes represent users, and edges represent connections or interactions between them.

3. **Define Betweenness.**
    - Betweenness is a measure in graph theory quantifying the number of shortest paths passing through a node, indicating its importance in connecting other nodes in the network.
4. **What is Clique Percolation Method (CPM)?**
    - CPM is a technique used to detect overlapping communities in networks by identifying cliques (complete subgraphs) that share common nodes and percolating them to form larger communities.
5. **What is graph partitioning?**
    - Graph partitioning involves dividing a graph into disjoint subsets (partitions) to enhance analysis or optimize performance, often used in parallel computing or network optimization.
6. **How to find overlapping communities in a social network graph?**
    - Overlapping communities in social network graphs can be identified using algorithms that detect nodes belonging to multiple communities based on their connectivity patterns or structural roles.
7. **What are the different approaches to discover overlapping communities?**
    - Approaches include clique-based methods, modularity optimization, and algorithms based on node similarity or link prediction, each targeting specific characteristics of overlapping communities.
8. **What is online advertising?**
    - Online advertising refers to promoting products or services through internet-based platforms such as websites, search engines, and social media, allowing advertisers to reach targeted audiences efficiently.
9. **What are the issues in online advertising?**
    - Challenges include ad fraud, ad blocking, privacy concerns, targeting accuracy, and measuring ad effectiveness, impacting the efficiency and reliability of online advertising campaigns.
10. **What is the Adward problem?**
    - The "Adward" problem is not a recognized term in the context of online advertising. If referring to a specific issue or challenge, further clarification may be needed.
11. **How do you mine social network news feeds?**
    - Mining social network news feeds involves analyzing user interactions, content preferences, and network structures to extract insights, trends, and sentiments, aiding in content recommendation and user engagement.
12. **What is web mining?**
    - Web mining is the process of extracting useful information, patterns, or knowledge from web data, including web pages, links, and user interactions, to support various applications such as search engine optimization and data analytics.